

_____ 33. With regard to the oral cavity, which statement is true?

- a. Sebaceous and eccrine sweat glands are prominent in the transitional [vermillion] zone of a lip.
- b. The internal region of the lip is covered by a stratified squamous keratinized epithelium.
- c. In the submucosa of the cheeks are minor salivary glands and occasional sebaceous glands (Fordyce spots).
- d. Pseudostratified columnar ciliated epithelium covers the oral surface of the soft palate.
- e. The palatal raphe is a median fold that joins the internal aspect of each lip to the gum.

_____ 34. Concerning the histology of the teeth, which statement is true?

- a. The highly calcified matrix of enamel is composed of type III collagen fibrils.
- b. The organic matrix of enamel is secreted by odontoblasts.
- c. Dentin is a calcified connective tissue that is harder than bone.
- d. Ameloblasts are tall, columnar epithelial cells that line the pulp chamber.
- e. Like bone, cementum is perforated by Sharpey's fibres and neurovascular elements.

_____ 35. With regard to the tongue, which statement is false?

- a. The filiform papillae are the smallest and most abundant papillae on the dorsum of the tongue.
- b. Filiform papillae are the only papillae on the tongue which do not contain taste buds.
- c. Foliate papillae are well developed in man and are seen on the edges of the tongue arranged in a "V" shape.
- d. Taste buds are found on circumvallate, fungiform, and foliate papillae.
- e. Fungiform papillae are mushroom shaped papillae on the tongue which contain taste buds.

_____ 36. Which organ has a mucosa lined by nonkeratinized stratified squamous epithelium?

- a. Esophagus
- b. Stomach
- c. Small intestine
- d. Large intestine
- e. Rectum

_____ 37. Concerning the stomach, which statement is true?

- a. It is specifically responsible for the absorption of vitamin B₁₂ and the reabsorption of conjugated bile salts.
- b. Goblet cells are more abundant to produce mucus as well as appreciable amounts of lysozyme.
- c. Hydrochloric acid and intrinsic factors are secreted by enteroendocrine cells of cardia and pylorus.
- d. Parietal cells also called oxyntic cells contain mucinogen granules and pepsinogen.
- e. The folds of the empty stomach are called rugae.

_____ 38. With regard to the small intestine, which statement is true?

- a. The muscularis externa in the upper third of duodenum and jejunum is composed of striated muscle.
- b. Crypts of Lieberkühn are lymphatic capillaries within a villus of the small intestine.
- c. Microvilli that are seen on the epithelial cells in the small intestine form the brush border or

striated border.

d. The plicae circulares are projections with a core of mucosa. e. Villi are finger like projections of submucosa.

_____ 39. With regard to the large intestine, which statement is false?

- a. The Paneth cells of the submucosal glands house bacteria that produce vitamin B₁₂ and vitamin K.
- b. Lacks villi and contains a disproportionately large number of goblet cells.
- c. The lamina propria has many macrophages, plasma cells, eosinophils, and lymphoid nodules.
- d. The muscularis externa has three thick bands of muscle called the tenia coli.
- e. The large intestine is shorter than the small intestine but has a wide lumen.

_____ 40. With regard to the pancreas, which statement is false?

- a. Although is a mixed exocrine and endocrine gland, the majority of the secretory cells are exocrine cells.
- b. The myoepithelial cells of the striated ducts deliver bicarbonate ions into the primary secretion.
- c. The exocrine portion is a compound tubuloacinar gland, similar in structure to the parotid gland.
- d. Islets of Langerhans are incompletely delimited from the acini by Schwann cells.
- e. It is a lobulated retroperitoneal gland whose islet cells are compact mass of epithelial cells not cords or plates.

_____ 41. Which is not the function of the liver?

- a. Detoxification/removal of drugs and alcohol.
- b. Storage of calcium and bile.
- c. Storage of glycogen, fat soluble vitamins, iron and cholesterol.
- d. Production of plasma proteins and lipoproteins.
- e. Fetal RBC production

_____ 42. Concerning the Kupffer cells, which statement is false?

- a. uptake, store, and release of retinoids.
- b. are found on the luminal surface of the endothelial cells, within the sinusoids, but lack basal lamina.
- c. metabolize aged erythrocytes and they digest hemoglobin.
- d. are derived from monocytes and they destroy bacteria that enter the portal blood through the large intestine.
- e. Most of them are located in the periportal region of the liver lobule, where they are very active in phagocytosis.

_____ 43. Concerning the GI tract, select the statement which is not correct.

- a. The mucosa of the gallbladder is made of simple columnar epithelium.
- b. The masticatory mucosa is located on the hard palate and gingiva.
- c. The adventitia consists of only a single layer of epithelial cells and underlying connective

tissue.

- d. The muscularis externa consists of Auerbach's plexus.
- e. Brünner's glands in submucosa of the duodenum produce mucin containing alkaline fluid and the polypeptide hormone urogastrone.

_____44. With regard to the epidermis, which statement is false?

- a. The stratum lucidum is present only in thick skin.
- b. Langerhans cells are macrophages seen in the skin.
- c. The keratinocyte is the most abundant cell in the epidermis.
- d. Fibroblasts form stratified squamous epithelium of the epidermis.
- e. The cells in the stratum granulosum contain keratohyaline granules.

_____45. What type of glands are the ceruminous glands?

- a. Sebaceous glands
- b. Eccrine sweat gland
- c. Endocrine gland
- d. Apocrine sweat gland
- e. Oil gland

_____46. The most common cells in dermis are:

- a. Keratinocytes.
- b. Melanocytes.
- c. Langerhans cells.
- d. Fibroblasts.
- e. Merkel cells.

_____47. Which of the following is not a function of the kidney?

- a. Erythropoietin production
- b. Vitamin D modification
- c. Acid base balance
- d. Aldosterone production
- e. Renin production

_____48. What vessels arise from the efferent arterioles?

- a. Afferent arteriole
- b. Efferent arteriole
- c. Capillary tuft
- d. Peritubular capillaries
- e. Vasa recta

_____49. Select the correct statement pertaining the morphology of the kidney.

- a. The distal convoluted tubules exhibit brush borders and apical canaliculi.
- b. The renal pyramid and its associated cortex is referred to as the renal lobe.
- c. Macula densa cells are located in the urinary pole of the nephrons.
- d. Renin is secreted from the visceral layer of Bowman's capsule referred to as area cribrosa.
- e. Interstitial cells in the renal medulla secrete erythropoietin through the filtration slits.

_____50. In the urinary system, which statement is false?

- a. The ureter is lined by transitional epithelium and it does not contain lymphatics.
- b. Polyploidy and binucleated cells are common among cells of the transitional epithelium.
- c. Epithelial cells lining the thin limb of the loop of Henle are simple squamous.
- d. Mesangial cells are modified smooth muscle cells responsible for the secretion of rennin
- e. Aldosterone acts on the distal convoluted duct.

II. EMBRYology

_____ 43. During the development of the digestive system, which statement is true?

- a) omphalocele is a persistence of the herniation of abdominal contents into the proximal part of the umbilical cord.
- b) the superior two-thirds of the adult anal canal are derived from proctodeum.
- c) the junction of the epithelium is derived from the ectoderm of the proctodeum.
- d) the inferior one-third of the anal canal is supplied with superior rectal artery.
- e) the anal membrane normally breaks down by the end of the second trimester..

_____ 44. Concerning the development of the GIT and associated structures, which statement is true?

- a) The midgut begins with $\frac{1}{2}$ of the duodenum and ends with $\frac{1}{3}$ of the sigmoid colon.
- b) Stomach rotates 270° counterclockwise, similar to the midgut.
- c) The middle of the duodenum would receive blood from celiac and superior mesenteric artery.
- d) The hepatocytes of liver are of mesodermal in origin, while the stroma is of endodermal origin.
- e) The exocrine portion of pancreas develops from endoderm of the gut, while the endocrine portion develops from the local mesoderm.

_____ 45. With regard to the development of the digestive system, select the correct statement.

- a) the foregut may give rise to the lower respiratory system.
- b) the midgut may give rise to the descending colon and sigmoid colon.
- c) the hindgut derivatives are supplied by inferior mesenteric artery.
- d) during the rotation of the stomach, the lesser curvature moves to the left and the greater curvature moves to the right.
- e) the dorsal pancreatic bud forms the uncinat process and part of the head of the pancreas.

_____ 46. A three-week old male infant comes into the Emergency Room for weight loss and acute dehydration. The infant has been projectile vomiting and the condition is worsening. Which of the following is most likely malformation?

a) Gastric atresia	b) Pyloric stenosis	c) Annular pancreas
d) Malrotation of the gut	e) Intestinal duplication	

_____ 47. Concerning the development of the urinary system, which statement is false?

- a) Urinary system begins to develop before the genital system.
- b) The metanephric diverticulum is the primordium of the ureter.
- c) Nephrons are derived from the metanephric mass of intermediate mesoderm.
- d) Differentiation of the nephrons depends upon induction by the collecting tubules.
- e) The kidneys attain their definitive adult position toward the end of the fetal period.

_____ 48. Concerning the development of the urinary bladder and urethra, which statement is false?

- a) The female urethra is derived from endoderm of the urogenital sinus.
- b) In adults, the urachus is represented by the median umbilical ligament.
- c) The urinary bladder becomes a pelvic organ during the 9th week of development.

- d) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
- e) The epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.

_____ 49. With regard to the *congenital anomalies of the* kidneys and ureters which statement is *false*?

- a) Accessory renal arteries are end arteries.
- b) The “ascent” of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.
- c) Incontinence is a common complaint resulting from an ectopic ureteric orifice in both sexes.
- d) Unilateral renal agenesis is more common in males than females.
- e) None of the above.

_____ 50. During the development of the Urinary System, which statement is true?

- a) The urorectal septum divides the urachus from the allantois.
- b) Bilateral renal agenesis is associated with oligohydramnios.
- c) The podocytes of Bowman’s capsule are derived from endoderm and ectoderm.
- d) Urachal cyst is a remnant of the cloaca.
- e) Urine production begins shortly after birth.

HISTOLOGY

1. Concerning the *teeth and associated tissues*?

- a) Dentin is more calcified than enamel.
- b) Pulp is dense connective tissue.
- c) Enamel is replaced in the root by cementum.
- d) Odontoblasts make enamel.
- e) Periodontal membrane is loose connective tissue.

2. Which of the following types of lingual papillae lack taste buds?

- a) circumvallate
- b) filiform
- c) foliate
- d) fungiform
- e) none

3. Serous demilunes,

- a) produce IgA
- b) border directly on salivary intercalated ducts
- c) produce mucinogen
- d) produce lysozyme
- e) occur only in parotid glands

1. In the wall of the stomach,

- a) simple squamous epithelium covers its outer surface.
- b) forms temporary folds called plicae circulares when the stomach is empty.
- c) stratified squamous epithelium covers its inner surface.
- d) three layers of skeletal muscle comprise its muscularis externa
- e) lacks a definitive submucosa.

—————20. Regarding the *stomach*:

- a) Is able to withstand the acidic environment because it is lined by stratified squamous keratinized epithelium.
- b) Has its gastric glands best developed in the pylorus than the other regions.
- c) Has chief cells that are intensely eosinophilic.
- d) Has its muscularis externa made by skeletal muscle fibres.
- e) Has parietal cells in its glands that secrete HCl.

20. Pernicious anemia can result from insufficient intrinsic factor secretion by which of the following cells?

- a) Paneth cells
- b) enterocytes
- c) ameloblasts
- d) oxyntic (parietal) cells
- e) zymogenic (chief) cells

56. Mucosal glands of the pylorus,

- a) resemble those in the body of the stomach.
- b) secrete more pepsinogen than do fundic glands.
- c) have shallower pits than those in the fundic region.
- d) secrete mainly mucus.
- e) contain abundant parietal cells in their basal half.

20. Which one of the following marks best the ileum from the duodenum and jejunum?

- a) Long and broad leaf like villi.
- b) Absence of taenia coli.
- c) Peyer's patches.
- d) Plicae circulares.
- e) Well developed intestinal glands.

—————20. Which of the following organs characteristically contains the most goblet cells in the mucosa covering its villi?

- a) duodenum
- b) ileum
- c) colon
- d) cecum
- e) stomach

3. Vermiform appendix,

- a) is a blind-ended evagination of the ileum.
- b) has few lymphoid nodules in the lamina propria and submucosa.
- c) has relatively shallow intestinal glands.
- d) longitudinal smooth muscle gathers into its teniae coli.
- e) contains short villi.

———51. Hepatic sinusoids,

- a) are bordered directly by hepatocytes.
- b) contents flow toward the portal triads.
- c) are surrounded by the space of Disse.
- d) lumen is entirely sealed by junctional complexes.
- e) contents empty into a cholangiole (canal of Herring).

———20. Skeletal muscle may occur in the muscularis externa at the junction between which of the following pairs of organs?

- a) pharynx and esophagus
- b) esophagus and stomach
- c) stomach and duodenum
- d) duodenum and jejunum
- e) ileum and cecum

———20. Diarrhea may result if which of the following organs fails to carry out its primary role in absorbing water from the feces?

- a) anal canal
- b) colon
- c) cecum
- d) small intestine
- e) stomach

———54. Which of the following statements is correct regarding the *intestine*?

- a) Intestinal glands in the large intestine are branched alveolar type.
- b) The striated border of the small intestine is made by microvilli.
- c) Small intestine has taenia coli.
- d) Large intestine has villi.
- e) None of the above.

———20. Myoepithelial cells,

- a) are usually found outside the acinar basal lamina
- b) can not contract
- c) contain actin and myosin in their cytoplasm
- d) are found around the bile canaliculi.
- e) are responsible for emptying the gallbladder.

———20. Which of the following glands lacks serous acini but contains serous demilunes?

- a) liver
- b) parotid gland
- c) pancreas
- d) sublingual gland
- e) submandibular gland.

———20. Regarding the gastric glands:

- a) Are best developed in the pylorus than in the other regions of stomach.
- b) Has no submucosa.
- c) Those in the pyloric region, have short pit, but long glands.
- d) Are simple branched in type.
- e) Those in the cardiac region have more type of secretory cells than those of fundus.

———20. The centroacinar cells of pancreas are:

- a) Larger in size than the secretory cells.
- b) More eosinophilic than the secretory cells.

- c) Are found in the middle of the Islet of Langerhans. d) More basophilic than the secretory cells.
e) Make the beginning of the intercalated ducts.

_____20. Which of the following is not a component of the *portal triad of the liver*?

- a) Branch of hepatic artery b) Branch of portal vein
c) Bile duct d) Central vein of liver e) None of the above

Parotid *glands* produce purely serous secretion in the adult.

_____74. Which of the following is a factor produced or activated in the dermis that stimulates calcium uptake in the digestive tract?

- a) calcitonin b) intrinsic factor c) melanin d) vitamin A e) vitamin D

_____57. Permanent circular folds (plicae circulares) is a characteristic structural feature of the walls of which of the following organs?

- a) anal canal b) colon c) cecum d) small intestine e) stomach

2. The absorptive cells of the small intestine,

- a) are also called enteroendocrine cells. b) many microvilli cover their basal surfaces.
c) absorb lipids by active transport. d) use absorbed lipids to synthesize triglycerides.
e) undergo mitosis at the tips of the villi and are sloughed into the crypts.

21. Brunner's glands,

- a) are characteristic components of the jejunal wall. b) produce serous secretion that is rich in digestive enzymes.
c) lie in the submucosal layer. d) empty their secretions into the lacteals.
e) are composed of collections of goblet cells.

14. In the digestive system,

- a) the colon has more goblet cells but lacks villi.
-----b) the space of Disse is bordered directly by hepatocytes.
-----c) the exocrine pancreas contain centoacinar cells.
-----d) the hepatic portal vein carry blood rich in bilirubin.
-----e) cholecystokinin stimulates the relaxation of smooth muscle in the gallbladder.

15. With regard to the digestive system,

- a) central vein is located at the center of a classic liver lobule.
-----b) albumin is an endocrine secretory product of the liver.
-----c) hepatocytes exhibit both exocrine and endocrine secretory activity.
-----d) ascending colon has both adventitial and serosal coverings.
-----e) Paneth cell secretes lysozyme.

_____53. Regarding the *stomach*:

- a) Has its muscularis externa made by skeletal muscle fibres.
b) Has its gastric glands best developed in the pylorus than the other regions.
c) Is able to withstand the acidic environment because it is lined by stratified squamous keratinized epithelium.
d) Has parietal cells in its glands that secrete HCl.
e) Has chief cells that are intensely eosinophilic

_____55. Which one of the following marks best *ileum* from *duodenum* and *jejunum*?

- a) Long and finger like villi.
- b) Well developed Peyer's patches.
- c) Well developed plicae circularis.
- d) Well developed microvilli.
- e) Simple columnar lining epithelium.

_____60. Which of the following glands lacks serous acini but contains serous demilunes?

- a) liver
- b) parotid gland
- c) pancreas
- d) sublingual gland
- e) submandibular gland

_____63. Vermiform appendix,

- a) is a blind-ended evagination of the ileum.
- b) has few lymphoid nodules in its walls.
- c) has relatively shallow intestinal glands.
- d) contains short villi.
- e) longitudinal smooth muscle gathers into its teniae coli.

72. Which of the following components of the respiratory tract wall increase(s) in amount from trachea to alveoli?

- a) cilia
- b) elastic fibers
- c) smooth muscle
- d) cartilage
- e) goblet cells

2. Which cell is a respiratory macrophage?

- a. Kupffer cell
- b. Histiocyte
- c. Dust cell or alveolar macrophage
- d. Langerhans cell
- e. Microglia

7. What type of epithelium lines the trachea?

- a. Simple squamous epithelium
- b. Simple cuboidal epithelium
- c. Simple columnar epithelium
- d. Stratified squamous epithelium
- e. Pseudostratified epithelium

17. Pulmonary surfactant,

- a) is secreted by type I pneumocytes.
- b) consists of a monolayer of phospholipids overlying an aqueous hypophase.
- c) prevents alveolar collapse by increasing surface tension.
- d) may promote the growth of bacteria.
- e) excessive production in the lungs of premature infants causes respiratory distress syndrome.

3. What type of cells are found in the olfactory mucosa?

- a. Basal cells
- b. Brush cells
- c. Olfactory cells
- d. Sustentacular cells
- e. All of the above

Larynx is partly lined by nonkeratinized stratified squamous epithelium. The epiglottis is composed of elastic cartilage.

Bronchioles do not have cartilage and submucosal glands.

The vestibule is lined by stratified squamous epithelium.

There is no fibrocartilage in the larynx.

The rings of the trachea are composed of **C-shaped** hyaline cartilage.

The pharynx is lined by both stratified squamous epithelium and ciliated pseudostratified epithelium with goblet cells.

An olfactory cell is a bipolar neuron.

The Clara cell is found in the terminal bronchioles.

The paranasal sinuses are lined by ciliated pseudostratified epithelium with goblet cells.

gas exchange takes place in the terminal bronchiole. FFFFFF

The epiglottis is elastic cartilage.

5. Which cell is a squamous pulmonary epithelial cell?

- a. Clara cell b. Type I pneumocyte c. Type II pneumocyte d.
Dust cell e. Brush cell

2. Which cell is found in large numbers in the terminal bronchioles?

- a. Clara cell b. Type I pneumocyte c. Type II pneumocyte d.
Dust cell e. Brush cell

5. What type of epithelium is found in the respiratory mucosa of man?

- a. Non-ciliated pseudostratified columnar epithelium with goblet cells
b. Ciliated pseudostratified columnar epithelium with goblet cells
c. Simple columnar epithelium d. Stratified squamous epithelium e.
Transitional epithelium

7. What type of tissue forms the alveoli in the lung?

- a. Simple squamous epithelium b. Simple cuboidal epithelium
c. Simple columnar epithelium
d. Stratified squamous epithelium e. Pseudostratified epithelium

10. Which cell secretes surfactant?

- a. Clara cell b. Type I pneumocyte c. Type II pneumocyte d.
Dust cell e. Brush cell

14. Pneumocytes type I:

- a) Are more in number than alveolocytes type II. b) Could give rise to pneumocytes type II.
c) Produce surfactant. d) Have large round centrally located nuclei.
e) Line more surface area of the alveolar wall than do the alveolocytes type II.

15. Bronchiole that contains *pseudostratified columnar* type of epithelium is:

- a) Primary bronchiole. b) Respiratory bronchiole. c) Terminal bronchiole.
d) All of the above. e) None of the above.

16. Which of the following is missing from the wall of the terminal bronchioles?

- a) Cartilage b) Smooth Muscles c) Lamina propria
d) Clara cells e) All of the above.

20. Bronchioles:

- a) Have well developed cartilages
- b) Are lined by stratified squamous epithelium
- c) Have no muscle
- d) Have clara cells
- e) Have mucous producing glands

16. The visceral pleura,

- a) can be described as an adventitia.
- b) is devoid of lymphatic vessels.
- c) includes a layer of simple squamous epithelium.
- d) lacks elastic fibers.
- e) covers the outer wall of the pleural cavity.

18. In the respiratory system,

- a) respiratory bronchioles have an epithelial lining in which cilia are typically present but goblet cells are absent.
- b) the left lung contains vertical primary bronchus.
- c) alveoli occur in clusters at the end of the bronchial tree.
- d) a smoker's respiratory epithelium typically develops more goblet cells.
- e) the lamina propria contains mucous glands at the level of the alveolar ducts.

———70. Pulmonary surfactant,

- a) is secreted by type I pneumocytes.
- b) may promote the growth of bacteria.
- c) consists of a monolayer of phospholipids overlying an aqueous hypophase.
- d) prevents alveolar collapse by increasing surface tension.
- e) excessive production in the lungs of premature infants causes respiratory distress syndrome.

URINARY

———75. Which of the following is the modified smooth muscle cell responsible for the secretion of rennin?

- a) juxtaglomerular cell
- b) mesangial cell
- c) podocyte
- d) polkissen (extraglomerular mesangial cell)
- e) macula densa cells

———82. Which of the following is not the function of *mesangial cells*?

- a. synthesis of extracellular matrix.
- b. endocytosis and contractility.
- c. maintenance of acid-base balance in the blood.
- d. structural support to the glomerulus.
- e. increase blood flow and the effective surface area available for filtration.

———76. What is a renal pyramid and its associated cortex referred to?

- a. Medulla
- b. Lobe
- c. Renal columns
- d. Nephron
- e. Medullary ray

———78. Which of the following cell types comprises the visceral layer of Bowman's capsule?

- a) endothelial cell
- b) juxtaglomerular cell
- c) mesangial cell
- d) podocyte
- e) polkissen (extraglomerular mesangial cell)

37. Cortex of the kidney contains the following structures *except*

- a. proximal convoluted tubule (PCT)
- c. renal corpuscle
- e. None of them

- b. thin segment of loop of Henle
- d. distal convoluted tubule (DCT)

———79. Which of the following is characteristic of the proximal but not of the distal convoluted tubules?

- a) convoluted portion lies in the renal cortex
- b) epithelial lining is similar to that of the loop of Henle
- c) exhibits a brush border
- d) lining cells are basophilic
- e) secretes glucose and amino acids into the filtrate

———80. Which of the following is not a function of the kidney?

- a. Erythropoietin production
- b. Vitamin D modification
- c. Acid base balance
- d. Aldosterone production
- e. Renin production

———81. Select the *correct statement* pertaining the *morphology of the kidney*.

- a. The peritubular capillaries supply blood to the glomeruli.
- b. The efferent arteriole branches off the interlobular artery.
- c. Renin is secreted from the visceral layer of Bowman's capsule.
- d. The capillary tuft of the nephron is referred to as the vasa recta.
- e. The renal pyramid and its associated cortex is referred to as the renal lobe.

———83. In the *urinary system*, which statement is *false*?

- a. The ureter is lined by transitional epithelium and it does not contain lymphatics.
- b. Polyploidy and binucleated cells are common among cells of the transitional epithelium.
- c. Distal convoluted tubule is longer and more convoluted than proximal convoluted tubule.
- d. Calyces are lined by transitional epithelium.
- e. Aldosterone acts on the distal convoluted duct.

———84. In the *urinary system*, which statement is *true*?

- a. Interstitial cells in the renal medulla secrete erythropoietin.
- b. Macula densa are located in the urinary pole of the nephrons.
- c. The male urethra contains transitional epithelium throughout its length.
- d. Endothelial cells of glomerular capillaries are of non-fenestrated variety.
- e. The distal convoluted tubules possess neither brush borders nor apical canaliculi.

42. Which of the following statements is true about the urinary system?

- a. Renin is secreted from the visceral layer of Bowman's capsule
- b. The glomerulus and Bowman's capsule make up the renal corpuscle.
- c. The parietal layer of Bowman's capsule consists of modified smooth muscle cells called podocytes.
- d. The lining epithelium of the female urethra is simple ciliated columnar and transitional epithelium.
- e. The juxtaglomerular cells are specialized cells in the distal convoluted tubule that are sensitive to sodium.

43. In the urinary system, which statement is false?

- a) mesangial cells clear occasional build-up of cellular debris and some antigen-antibody complexes by phagocytic activity.
- b) pseudostratified columnar epithelium lines the prostatic urethra.
- c) Composition of a renal lobule is best described by medullary ray and all nephrons that empty into it.
- d) Collections of cortical tissue between the medullary pyramids are called renal column of Bertin.
- e) Arcuate arteries and veins are typically seen at the border between the renal cortex and medulla.

44. In the urinary system, which statement is false?

- a. Aldosterone acts on the distal convoluted duct.
- b. The ureter is lined by transitional epithelium and it does not contain lymphatics.
- c. Polyploidy and binucleated cells are common among cells of the transitional epithelium.
- d. DCT is longer and more convoluted than PCT.
- e. Calyces are lined by transitional epithelium.

45. In the urinary system, which statement is true?

- a. the male urethra contains transitional epithelium throughout its length.
- b. the distal convoluted tubules possess neither brush borders nor apical canaliculi.
- c. macula densa are located in the urinary pole of the nephrons.
- d. endothelial cells of glomerular capillaries are of non-fenestrated variety.
- e. interstitial cells in the renal medulla secrete erythropoietin.

_____ 11. With regard to the *urinary system*, which statement is true?

- a) Podocytes are phagocytic cells and elaborate extracellular matrix that form the visceral layer of the Bowman's capsule.
- b) The proximal convoluted tubule lacks a brush border.
- c) The cortical collecting tubules are located primarily at the apex of the renal pyramids and are lined by pseudostratified ciliated columnar cells.
- d) The transitional epithelium of a relaxed urinary bladder has rounded superficial cells that bulge into the lumen.
- e) The female urethra is lined primarily by stratified squamous nonkeratinized epithelium, although patches of transitional epithelia are present.

_____ 11. In the *urinary system*, which statement is false?

- a) A nephron and its collecting tubules constitute a uriniferous tubule.
- b) The interstitial cells in the renal medulla secrete vasodepressor substances.
- c) Juxtaglomerular cells are modified smooth muscle cells that release renin.
- d) Juxtaglomerular apparatus is located at the vascular pole of the renal corpuscle.
- e) Extraglomerular mesangial cells are tall, narrow, closely packed epithelial cells of the distal tubule.

_____ 11. Which of the following statements is false concerning the urinary system?

- a. The renal cortical tissue that extends between the pyramids is called renal column of Bertin.

- b. A pyramid and the cortical tissue over it is referred to as a renal lobe.
- c. The PCT can be identified in histological section by the presence of acidophilic cytoplasm of the lining cuboidal cells.
- d. Afferent arterioles of the glomerulus are branches of arcuate artery.
- e. Antidiuretic hormone makes the epithelium of collecting duct permeable to water.

38. In the urinary system,

- a) juxtaglomerular cells are modified smooth muscle cells that release renin.
- b) the phagocytic mesangial cells are present in the interstitium of the renal glomeruli.
- c) interstitial cells in the renal medulla secrete vasodepressor substances.
- d) the ascending thick limb of the loop of Henle is impermeable to water even in the presence of ADH.
- e) the male urethra contains transitional epithelium throughout its length.

39. Concerning the urinary system,

- a) the distal convoluted tubules have no brush borders and apical canaliculi.
- b) macula densa are located in the urinary pole of the nephrons.
- c) podocytes are highly modified epithelial cells that form the visceral layer of Bowman's capsule.
- d) papillary collecting tubules empty at area cribosa.
- e) the female ureter is lined primarily by stratified squamous epithelium.

Large collecting tubules near the apex of the pyramid are the ducts of Bellini or papillary ducts.

MALE

—————105. *Testis* has the following coverings except:

- a. tunica adventitia
- b. tunica vasculosa
- c. tunica albuginea
- d. tunica vaginalis
- e. None

—————106. In the *testis*, which statement is *true*?

- a. Interstitial cells (of Leydig) produce testicular fluid.
- b. The seminiferous tubules are lined with a complex stratified epithelium.
- c. The secondary spermatocytes are the largest cells of the spermatogenetic lineage.
- d. The tunica vaginalis is thickened on the posterior surface to form the mediastinum testis.
- e. During spermiogenesis, the spermatids quickly undergo the second meiotic division to give spermatozoa.

—————107. Which of the following is NOT considered part of the male genital duct system?

- a. Rete testis
- b. Tubuli recti
- c. Seminal vesicle
- d. Ductus deferens
- e. Ductus epididymidis

—————108. Which statement is *false* with regard to the *prostate gland*?

- a. It produces testicular androgens.
- b. It is composed of compound tubuloalveolar glands.
- c. It is the largest accessory structure of the male reproductive system.
- d. The glands of the prostate lie in a fibromuscular stroma, also referred to as a fibroelastic stroma.

e. Corpora amylacea {prostatic concretions} are the spherical structures seen in some prostatic alveoli.

_____ 109. Which of the following statements is *false* in the *male reproductive system*?

- a. Tunica albuginea is the thickest in epididymis.
- b. The subcutaneous tissue in penis is devoid of fat.
- c. The secretion of seminal vesicle is particularly rich in fructose.
- d. Secondary spermatocytes are smaller than the primary spermatocytes.
- e. The epithelium lining the ductus epididymis is pseudostratified columnar.

47. Which cells produce testosterone?

- a. Interstitial cells of Leydig
- b. Juxtaglomerular cells
- c. Sertoli cells
- d. Sustentacular cells
- e. Spermatogonia

48. In the testis, which statement is true?

- a. interstitial cells (of Leydig) produce testicular fluid.
- b. the seminiferous tubules are lined with a complex stratified epithelium.
- c. the secondary spermatocytes are the largest cells of the spermatogenetic lineage.
- d. the tunica vaginalis is thickened on the posterior surface to form the mediastinum testis.
- e. during spermiogenesis, the spermatids quickly undergo the second meiotic division to give spermatozoa.

51. Which of the following statements is false in the male reproductive system?

- a. The secretion of seminal vesicle is particularly rich in fructose.
- b. Secondary spermatocytes are smaller than primary spermatocytes.
- c. The epithelium lining the ductus epididymis is pseudostratified columnar.
- d. Tunica albuginea is the thickest in epididymis.
- e. The subcutaneous tissue in penis is devoid of fat

_____ 11. Select the correct statement with regard to the male reproductive system.

- a. Interstitial cells of Leydig produce testosterone.
- b. Sertoli cells are also called sustentacular cells or nurse cells.
- c. Sertoli cells produce inhibin and androgen-binding protein.
- d. Sertoli cells are directly involved in spermatogenesis.
- e. None of the above.

_____ 11. Concerning the male accessory glands, which statement is false?

- a. Pacinian corpuscles are spherical structures seen in some prostatic alveoli.
- b. Pseudostratified epithelium lines the seminal vesicles.
- c. Pseudostratified epithelium lines the vas deferens.
- d. The prostate is made of compound tubuloalveolar glands.
- e. None of the above.

_____ 11. In the *male reproductive system*, which statement is false?

- a) Sertoli cells possess receptors for FSH.
- b) Ductus epididymidis secretes glycerophosphocholine.
- c) Ejaculatory duct has a very thick muscular wall.

d) The glans penis is the dilated distal end of corpus spongiosum _____ e)
None of the above.

_____ 11. In the *testes*, which statement is false?

- a) Tunica albuginea is the thick, fibrous connective tissue capsule of the testis.
- b) Interstitial cells of Leydig phagocytose excess cytoplasm discarded by maturing spermatids.
- c) Spermiogenesis is a unique process of cytodifferentiation whereby spermatids are transformed into spermatozoa.
- d) The critical testicular temperature is 35°C.
- e) None of the above.

35. Tunica albuginea is present in all, *except*

- a. penis
- b. epididymis
- c. testis
- d. ovary
- e. None of these

33. The secretion of seminal vesicle is rich in fructose

36. Secondary spermatocytes are smaller than primary spermatocytes. TRUE.

37. Type A spermatogonia are involved in spermatogenesis. FALSE.

38. Blood testis barrier is formed by Sertoli cells. TRUE.

39. Smooth muscle present in the ductus epididymis and ductus deferens, is richly innervated by parasympathetic fibres. FALSE.

40. In the epididymis, the height of the epithelium gradually decreases but the muscle gradually increases from head to tail. TRUE.

41. The subcutaneous tissue in penis is devoid of fat. TRUE.

42. The bulbourethral glands are present in the superficial perineal pouch. FALSE.

43. The urethra at the glans penis is lined by stratified squamous epithelium. TRUE.

44. Leydig cells have basophilic cytoplasm. FALSE.

45. Secondary spermatocytes are rarely encountered in section of testis. TRUE.

5. What is the surface modification seen on the cells of the epididymis?

- a. Microvilli
- b. Stereocilia
- c. Cilia
- d. Keratinization
- e. Plica circularis

7. Which of the following is true?

- a. There is one corpus cavernosa and one corpus spongiosum
- b. There is one corpus cavernosa and two corpora spongiosum
- c. There are two corpora cavernosa and one corpus spongiosum
- d. There are two corpora cavernosa and two corpora spongiosum
- e. There are four corpora cavernosa and two corpora spongiosum

8. What gland in the male is homologous to the greater vestibular gland in the female?

- a. Bartholin's gland
- b. Prostate gland
- c. Skene's gland
- d. Cowper's gland bulbourethral glands
- e. Brunner's gland

Pseudostratified epithelium lines the vas deferens.

Pseudostratified epithelium lines the seminal vesicles.

Pseudostratified epithelium lines the epididymis.

The prostate is made of compound tubuloalveolar glands.

The prostate is the largest accessory structure of the male reproductive system.

The glands of the prostate lie in a fibromuscular stroma. It is also referred to as a fibroelastic stroma. There are bundles of smooth muscle within dense irregular connective tissue.

The prostate is made of compound tubuloalveolar glands.

The glands of the prostate lie in a fibromuscular stroma. It is also referred to as a fibroelastic stroma. There are bundles of smooth muscle within dense irregular connective tissue.

The spherical structures seen in some prostatic alveoli are called prostatic concretions.

The male genital duct system contains: rete testes, tubuli recti, ductuli efferentes, ductus epididymidis, and ductus deferens.

Sertoli cells produce inhibin and androgen-binding protein. The interstitial cells of Leydig produce testosterone.

3. Which of the following is NOT considered an accessory gland?

a. Prostate

b. Bulbourethral gland

c. Seminal vesicles

d. Ductus deferens

e. None of the above

The tunica albuginea is the thick capsule around each testis.

FEMALE

—————110. Which of the following is true of the *ampulla* of the *uterine tube* but not of its *isthmus*?

a) Contains large mucosal folds that are extensively branched.

b) Contains smooth muscle in its walls. c) Is covered by serosa derived from the peritoneum.

d) Contains ciliated epithelial cells.

e) Contains mucus-secreting peg cells.

—————111. Which of the following statements is *true* about the *female reproductive system*?

a. Mucosa of the uterus is commonly called perimetrium.

b. Fornix of vagina is lined by stratified squamous epithelium.

c. The life span of corpus luteum of pregnancy is about 10 lunar months.

d. Primary oocyte completes its first meiotic division at the time of fertilization.

e. Nonciliated secretory cells present in the lining epithelium of uterine tube synthesise and store glycogen.

—————112. Which of the following statements is *true* in the *female reproductive system*?

a. The cells of theca interna secrete progesterone.

b. Germinal (surface) epithelium covering the ovary gives rise to ova.

c. Glassy membrane is the last structure to disappear in an atretic follicle.

d. The alkaline environment of vagina prevents growth of microorganisms.

e. The basic pH in the vaginal lumen is secretion of HCl by the tubuloalveolar glands of the vaginal walls.

52. Which of the following statements is true in the female reproductive system?

- a. Mucosa of the uterus is commonly called perimetrium.
- b. Fornix of vagina is lined by stratified squamous epithelium.
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- d. The alkaline environment of vagina prevents growth of microorganisms.
- e. the basic pH in the vaginal lumen is secretion of HCl by the tubuloalveolar glands of the vaginal walls.

_____ 11. What is the connective tissue which surrounds the ovary?

- a. Follicle b. Germinal epithelium c. Medulla d. Tunica albuginea e. Cortex

_____ 11. In the *female reproductive system*, which statement is true?

- a) The peg cells of the uterus secrete a nutrient-rich medium that nourishes the spermatozoa and preimplantation embryo.
- b) The basal layer of the endometrium is the thick smooth muscle tunic of the uterus.
- c) The inferior portion of the cervix, which is continuous with the lining of the vagina, is covered by stratified squamous keratinized epithelium.
- d) The epithelium of the vagina contains glycogen.
- e) Mammary glands are simple tubuloacinar glands.

46. Nonciliated secretory cells present in the lining epithelium of uterine tube are called peg cells.

47. The endometrical layer that is shed during each menstrual cycle is called stratum functionalis.

48. Granulosa cells that are adherent to the zona pellucida form corona radiata.

49. Mucosa of the uterus is commonly called endometrium.

50. Fornix of vagina is lined by stratified squamous epithelium.

51. Cervical cancer can be detected by a special cytological test called papanicolaou's test.

52. The superficial cells of the stratified squamous epithelium of vagina synthesise and store glycogen.

53. Corpus luteum is derived from theca interna and membrana granulosa of the ovarian follicle.

54. Ovulation occurs usually on the 14th day of a 28-day menstrual cycle.

55. Mammary gland is a modified apocrine sweat gland.

56. Primordial follicle is identified in section by the presence of
 a. zona pellucida b. single layer of flattened follicular cells c. theca folliculi
 d. antrum folliculi e. cumulus oophorus
58. Fertilization normally takes place at
 a. ampulla of uterine tube b. uterine cavity c. vagina d. cervical canal
59. Uterine tube is lined by simple columnar ciliated epithelium
60. Primary oocyte completes its first meiotic division at the time of ovulation
61. Secretory phase of endometrium is characterized by the presence of
 a. oedematous endometrium b. tortuous uterine glands c. secretion rich in glycogen
 d. all of the above
63. Lactating mammary gland presents all of the following features except that
 a. alveoli are large and branched b. alveoli contain secretory material
 c. connective tissue septae are thick and well developed d. intralobular ducts are few
64. Granulosa lutein cells of corpus luteum secrete progesterone
66. Germinal epithelium covering the ovary gives rise to ova. FALSE.
67. Glassy membrane is the last structure to disappear in an atretic follicle. TRUE.
68. Zona pellucida is derived from oocyte only. FALSE.
69. The cells of theca interna secrete progesterone. FALSE.
70. During pregnancy the myometrium undergoes hypertrophy. TRUE.
71. Straight arteries supply the superficial layer of the endometrium. FALSE.
72. Coiled arteries are under the influence of ovarian hormones. TRUE.
73. Proliferative phase of uterus coincides with the development of ovarian follicles. TRUE.
74. The alkaline environment of vagina prevents growth of micro organisms. FALSE.
75. Colostrum gives passive immunity to the newborn. TRUE.

- _____ 11. In the ovaries, which statement is true?
 a) The capsule contains tunica albuginea of dense irregular collagenous connective tissue.
 b) The primary follicles are dependent on FSH for their development.
 c) Corpus albicans is a temporary endocrine gland whose formation depends upon LH surge.
 d) Atretic follicles are found in the ovarian medulla. e) Theca interna cells manufacture androgens.

54. In the female reproductive system, which of the following statements is false?
 a. Ovulation is triggered by a dramatic increase in LH.
 b. During a pregnancy, hCG maintains the corpus luteum.
 c. Proliferative phase of uterus coincides with the development of ovarian follicles.
 d. In the ovaries, the medulla harbors most of the oocyte-containing ovarian follicles.
 e. during pregnancy the endometrium grows extensively by hypertrophy and hyperplasia.

55. Which of the following is true of the uterine tube's ampulla but not of its isthmus?

- a) contains mucus-secreting peg cells b) contains ciliated epithelial cells
c) is covered by serosa derived from the peritoneum d) contains smooth muscle in its walls
e) contains large mucosal folds that are extensively branched

56. Which of the following statements is false in the female reproductive system?

- a. Mammary gland is a modified apocrine sweat gland.
- b. The vagina lacks glands throughout its length and is lubricated by secretions from cervical glands.
- c. Lactating mammary gland secretes progesterone.
- d. Cervix does not participate in the endometrial (menstrual) phases.
- e. None of the above.

3. Which stage of the follicle is arrested in prophase?

- a. Primordial follicle b. Primary follicle c. Secondary follicle d. Mature follicle
e. Graafian follicle

4. What is the cavity within a secondary follicle?

- a. Graffian follicle b. Theca folliculi c. Granulosa cells d. Zona pellucida e. Antrum

During the first week to 10 days, FSH is the main hormone which stimulates the growth of the follicles. Ovulation is triggered by a dramatic increase in LH.

During a pregnancy, HCG is the hormone which maintains the corpus luteum.

7. Which stage of the follicle is marked by the surrounding of flattened (squamous) follicular cells becoming cuboidal?

- a. Primordial follicle b. Primary follicle c. Secondary follicle d. Mature follicle
e. Graafian follicle

The connective tissue layer around the primary follicle is the theca folliculi.

The acidophilic glycoprotein coat surrounding the oocyte is the zona pellucida.

A cavity within the follicle is the antrum.

Underneath the germinal epithelium of the ovary is connective tissue called the tunica albuginea.

During a pregnancy, hCG is the hormone which maintains the corpus luteum.

12. Which follicular stage is also called an antral follicle?

- a. Primordial follicle b. Primary follicle c. Secondary follicle
d. Mature follicle e. Graffian follicle

2. What is the acidophilic glycoprotein coat which surrounds the oocyte?

- a. Graffian follicle b. Theca folliculi c. Granulosa cells d. Zona pellucida
e. Antrum

5. Which stage of the follicle is characterized by a surrounding of flattened (squamous) follicular cells?

- a. Primordial follicle
- b. Primary follicle
- c. Secondary follicle
- d. Mature follicle
- e. Graffian follicle

8.

What is the connective tissue layer around the primary follicle?

- a. Graffian follicle
- b. Theca folliculi
- c. Granulosa cells
- d. Zona pellucid
- e. Antrum

10. What is the connective tissue which surrounds the ovary?

- a. Follicle
- b. Germinal epithelium
- c. Medulla
- d. Tunica albuginea
- e. Cortex

_____ 1. Concerning *oogenesis*, which statement is true?

- a) All oogonia become primary oocytes during ovulation.
- b) Primary oocytes begin the first meiotic division before birth.
- c) Oogonia grow and give rise to secondary oocytes by mitotic division.
- d) Primary oocytes undergo cytodifferentiation to form three polar bodies prior to ovulation.
- e) The difference in sex chromosome complement of oocytes forms the basis of primary sex determination.

_____ 2. Concerning *spermatogenesis*, one statement is false.

- a) Type A spermatogonia undergo mitosis to provide a continuous supply of stem cells throughout the reproductive life of the male.
- b) Primary spermatocytes in males begin the first meiotic division before birth, but completion of prophase does not occur until adolescence.
- c) The primary spermatocytes are the largest germ cells in the seminiferous tubules.
- d) No primary spermatocytes form after birth.
- e) None of the above.

_____ 3. With regard to *spermatogenesis*, which statement is false?

- a. spermatids are transformed into spermatozoa by spermiogenesis.
- b. The entire process of spermatogenesis takes approximately 72 hours.
- c. Spermatogenesis is regulated by LH production by the pituitary.
- d. Spermatogenesis begins at puberty.
- e. Spermatogonia begin to increase in number at puberty.

_____ 4. With regard to *Sertoli cells*, which statement is false?

- a. They are involved in the regulation of spermatogenesis.
- b. They line the seminiferous tubules
- c. They support and nurture the male germ cells.
- d. They coagulate semen and form a vaginal plug
- e. They produce testicular fluid.

_____ 5. Regarding the *sperm transport*, which statement is false?

- a) Vesiculase coagulates some of the semen and forms a vaginal plug.

- b) Fructose in the semen, secreted by the seminal glands, is an energy source for the sperms.
- c) Spermatozoa are nonmotile during storage in the epididymis, but become motile in the ejaculate.
- d) Prostaglandins in the semen stimulate uterine motility. e) None of the above.

_____ 6. When does a *secondary oocyte* complete its second meiotic division to become a mature ovum?

- a) at ovulation
- b) before ovulation
- c) at fertilization
- d) at puberty
- e) before birth

_____ 7. Which statement is true about *female reproductive cycles*?

- a) LH stimulates the development of ovarian follicles and the production of estrogen by its follicular cells.
- b) ovulation is triggered by a surge of FSH production.
- c) if an oocyte is fertilized, the corpus luteum involutes and degenerates.
- d) ischemic phase occurs when the oocyte is fertilized. e) None of the above.

_____ 8. Concerning the *female reproductive system*,

- a) the usual site of fertilization is the ampulla of the uterine tube.
- b) the endometrium has a wall made up of a very thick smooth muscle layer.
- c) the basal layer of endometrium is sloughed off during each menstruation.
- d) the perimetrium is a trumpet-shaped opening towards the peritoneal cavity.
- e) the vaginal glands are sloughed off and discarded with each menstrual flow during pregnancy.

_____ 9. Which statement is false?

- a. The basal layer of the endometrium is sloughed off during menstruation.
- b. If a sperm penetrates the secondary oocyte, the second meiotic division is completed.
- c. The number of oocytes that ovulate is greatly reduced in women who take oral contraceptives.
- d. At ovulation, the nucleus of the secondary oocyte begins the second meiotic division, but progresses only to metaphase.
- e. Removal of the corpus luteum of pregnancy before the 4th month usually leads to abortion.

_____ 10. With respect to the *zona pellucida*, which statement is false?

- a. is composed of sulfated glycoproteins.
- b. prevents polyspermy
- c. prevents premature implantation
- d. functions as a porous filter
- e. arises by delamination from the embryoblast.

_____ 11. *Liquor folliculi*,

- a) stimulates the myoepithelial cells of the mature follicle to rupture.
- b) acts as a porous filter through which dissolved hormones and enzymes are prevented.
- c) causes the endometrial glands to secrete and prepare the endometrium for implantation.
- d) is composed of cumulus oophorus and polar bodies. e) None of the above.

45. Which of the following is not the function of *mesangial cells*?
- synthesis of extracellular matrix.
 - endocytosis and contractility.
 - maintenance of acid-base balance in the blood.
 - structural support to the glomerulus.
 - increase blood flow and the effective surface area available for filtration.
46. Which of the following is characteristic of the *proximal* but not of the *distal convoluted tubules*?
- exhibits a brush border.
 - secretes glucose and amino acids into the filtrate.
 - lining cells are basophilic.
 - convoluted portion lies in the renal cortex.
 - epithelial lining is similar to that of the loop of Henle.
47. Select the *correct statement* pertaining the *morphology of the kidney*.
- The peritubular capillaries supply blood to the glomeruli.
 - The efferent arteriole branches off the interlobular artery.
 - Renin is secreted from the visceral layer of Bowman's capsule.
 - The capillary tuft of the nephron is referred to as the vasa recta.
 - The renal pyramid and its associated cortex is referred to as the renal lobe.
48. In the *urinary system*, which statement is *false*?
- Calyces are lined by transitional epithelium.
 - Aldosterone acts on the distal convoluted duct.
 - The ureter is lined by transitional epithelium and it does not contain lymphatics.
 - Polyploidy and binucleated cells are common among cells of the transitional epithelium.
 - Distal convoluted tubule is longer and more convoluted than proximal convoluted tubule.
49. In the *urinary system*, which statement is *true*?
- Interstitial cells in the renal medulla secrete erythropoietin.
 - Macula densa are located in the urinary pole of the nephrons.
 - The male urethra contains transitional epithelium throughout its length.
 - Endothelial cells of glomerular capillaries are of non-fenestrated variety.
 - The distal convoluted tubules possess neither brush borders nor apical canaliculi.
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54. Which of the following is true of the *ampulla* of the *uterine tube* but not of its *isthmus*?

- a) Contains smooth muscle in its walls.
- b) Is covered by serosa derived from the peritoneum.
- c) Contains large mucosal folds that are extensively branched.
- d) Contains ciliated epithelial cells. e) Contains mucus-secreting peg cells.

55. Which of the following statements is *true* about the *female reproductive system*?

- a. Mucosa of the uterus is commonly called perimetrium.
- b. Fornix of vagina is lined by stratified squamous epithelium.
- c. The life span of corpus luteum of pregnancy is about 10 lunar months.
- d. Primary oocyte completes its first meiotic division at the time of fertilization.
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- b. Germinal (surface) epithelium covering the ovary gives rise to ova.
- c. Glassy membrane is the last structure to disappear in an atretic follicle.
- d. The alkaline environment of vagina prevents growth of microorganisms.
- e. The basic pH in the vaginal lumen is secretion of HCl by the tubuloalveolar glands of the vaginal walls.

57. Which of the following statements is *false* about the *sense organs*?

- a. There are more pain receptors than cold receptors in the body.
- b. Fingertips have more touch receptors than the skin of the back.
- c. End bulbs of Krause possess receptors for cold and pressure reception.
- d. Pacinian corpuscles are large, ellipsoid encapsulated, rapidly-adapting receptors.
- e. Meissner's corpuscles are unencapsulated, unmyelinated receptor terminals located in the deep dermis and hypodermis.

____ 78. Which of the following is not the adult derivative of the ureteric bud?
a) renal pelvis b) major calyx c) minor calyx d) collecting duct e) connecting tubule

____ 79. When does the metanephros become functional?
a) at week 3 of development b) at week 4 of development
c) at week 10 of development d) just before birth e) just after birth

____ 80. Urachal cyst is a remnant of
a) urogenital sinus b) urogenital ridge c) cloaca d) allantois e) mesonephric duct

____ 81. The transitional epithelium lining the urether is derived from
a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells

____ 82. The podocytes of Bowman's Capsule are derived from
a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells

____ 83. The proximal convoluted tubules of the definitive adult kidney are derived from
a) ureteric bud b) metanephric vesicle c) metanephric duct
d) pronephric tubules e) metanephric tubules

____ 84. A structure found within the adult female pelvis formed from the gubernaculum is
a) broad ligament b) superior ligament of the ovary c) medial umbilical ligament
d) round ligament of the uterus e) median umbilical ligament

____ 85. After the sinovaginal bulbs have proliferated and fused, they form a solid core of endodermal cells called
a) vestibule of the vagina b) uterovaginal primordium c) urogenital sinus
d) vaginal plate e) clitoris

____ 86. The labia minora arise embryologically from
a) phallus b) labioscrotal swellings c) sinovaginal bulbs
d) urogenital folds e) paramesonephric duct

____ 87. The uterine tubes of the adult female are derived embryologically from
a) mesonephric duct b) mesonephric tubules c) paramesonephric duct
d) paramesonephric tubules e) uterovaginal primordium

____ 88. The most common cause of female pseudointersexuality is
a) a 46, XO genotype b) a 47, XXY genotype
c) lack of androgen receptors d) congenital adrenal hyperplasia
e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)

____ 89. The most common cause of male pseudointersexuality is
a) a 46, XO genotype b) a 47, XXY genotype
c) lack of androgen receptors d) congenital adrenal hyperplasia
e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)

____ 90. The most common cause of testicular feminization syndrome is
a) a 46, XO genotype b) a 47, XXY genotype
c) lack of androgen receptors d) congenital adrenal hyperplasia
e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)

a) hypospadias b) epispadias c) cryptorchidism
d) hydrocele e) congenital inguinal hernia

a) testosterone
estrogen b) MIF c) testes-determining factor (TDF) d) progesterone e)

a) prochordal plate b) cloacal membrane c) primitive streak d) neural tube e) somites

a) skin plate b) eye c) primitive streak d) primitive node e) neural plate

a) neural plate b) cloacal membrane c) posterior
neuropore
d) primitive streak e) notochord

- a) Inguinal canals develop in both sexes.
- b) Sertoli cells constitute most of the seminiferous epithelium in the fetal testis.
- c) Because of the mode of descent of testis, the ductus deferens crosses anterior to the ureter.
- d) Gubernaculum does not play any role in the descent of ovaries.
- e) The persistent processus vaginalis in the female is called canal of nuck.

a) hypospadias b) epispadias c) cryptorchidism d) hydrocele e) congenital inguinal hernia

a) A dense tunica albuginea is the characteristic feature of testicular development in the fetus.

c) The fetal testis produces antimüllerian hormone by Sertoli cells at 6-7 weeks

e)

Round ligament of the uterus is formed from the gubernaculums. ttt

The Y chromosome carries a gene on its short arm that codes for testes-determining factor (TDF).ttt

____ 81. The transitional epithelium lining the urether is derived from
a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells

____ 82. The podocytes of Bowman's Capsule are derived from
a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells

____ 83. The proximal convoluted tubules of the definitive adult kidney are derived from
a) ureteric bud b) metanephric vesicle c) metanephric duct
d) pronephric tubules e) metanephric tubules

____ 84. A structure found within the adult female pelvis formed from the gubernaculum is
a) broad ligament b) superior ligament of the ovary c) medial umbilical ligament
d) round ligament of the uterus e) median umbilical ligament

85. After the sinovaginal bulbs have proliferated and fused, they form a solid core of endodermal cells called
a) vestibule of the vagina b) uterovaginal primordium c) urogenital sinus
d) vaginal plate e) clitoris

____ 86. The labia minora arise embryologically from
a) phallus b) labioscrotal swellings c) sinovaginal bulbs
d) urogenital folds e) paramesonephric duct

____ 87. The uterine tubes of the adult female are derived embryologically from
a) mesonephric duct b) mesonephric tubules c) paramesonephric duct
d) paramesonephric tubules e) uterovaginal primordium

____ 88. The most common cause of female pseudointersexuality is
a) a 46, XO genotype b) a 47, XXY genotype
c) lack of androgen receptors d) congenital adrenal hyperplasia
e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)

____ 89. The most common cause of male pseudointersexuality is
a) a 46, XO genotype b) a 47, XXY genotype
c) lack of androgen receptors d) congenital adrenal hyperplasia
e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)

22. Trigone on the posterior wall of the urinary bladder is formed by,
- a) incorporation of the lower end of the mesonephric ducts.
 - b) incorporation of the lower end of the pronephric ducts.
 - c) incorporation of the metanephric blastema.
 - d) incorporation of the mesonephric tubules.
 - e) incorporation of the pronephric tubules.